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B.Tech. Degree VI Semester Special Supplementary Examination January 2016

ME 1601 DYNAMICS OF MACHINERY (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

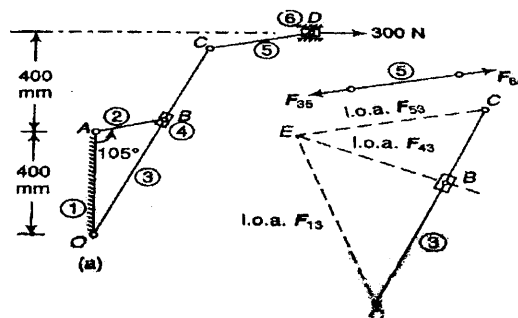
(8 × 5 = 40)

- I. (a) State and explain D' Alembert's principle.
 (b) What are the conditions for a body to be in equilibrium under the action of two forces, three forces and a torque?
 (c) Draw and explain the turning moment diagram of a four stroke IC engine.
 (d) What is meant by a self-locking brake?
 (e) Define the terms primary and secondary balancing as applied for balancing of reciprocating masses.
 (f) Explain briefly the method of balancing V-engines.
 (g) Explain the terms sensitiveness and hunting related to governors.
 (h) Derive the condition for transmitting maximum power in a flat belt drive.

PART B

(4 × 15 = 60)

- II. (a) Explain the effect of sliding friction in the static force analysis of slider crank mechanism. (5)
 (b) For the static equilibrium of the quick return mechanism shown in figure, determine the input torque T_2 to be applied on the link AB for a force of 300 N on the slider D. The dimensions of the various link are OA = 400 mm, AB = 200 mm, OC = 800 mm, CD = 300 mm. (10)



OR

- III. (a) Define and explain the term piston effort and crank pin effort. (5)
 (b) A horizontal steam engine running at 240 rpm has a bore of 20 cm and a stroke of 36 cm. The piston rod is 2 cm in diameter and connecting rod length is 90 cm. The mass of the reciprocating part is 7 kg and the frictional resistance is equivalent to a force of 500 N. Determine the following, when the crank is at 120° from the inner dead center, the mean pressure being $50 \times 10^2 \text{ N/m}^2$ on the cover side and $1 \times 10^2 \text{ N/m}^2$ on crank side. (10)
 (i) Thrust on the connecting rod (ii) Thrust on cylinder wall
 (iii) Load on the bearings (iv) Turning moment on the crank shaft.

(P.T.O.)

- IV. (a) Obtain expression for gyroscopic couple. (5)
 (b) A band and block brake has 14 blocks so that each block subtends an angle of 14° at the center of the rotating drum. The diameter of the drum is 750 mm and the thickness of the block is 65 mm. The two ends of the band are fixed to the pin on the lever at a distances of 50 mm and 210 mm from the fulcrum on the opposite sides. Determine the least force required to be applied at the lever at a distance of 600 mm from fulcrum if the power absorbed by the block is 180 kW at 175 rpm. Coefficient of friction between the block and drum is 0.35. (10)

OR

- V. (a) Explain the term spin axis and precession axis with the help of a neat sketch. (5)
 (b) The wheel of a motorcycle has a total moment of inertia of 2.5 kg-m^2 and the engine parts have a moment of inertia of 0.14 kg-m^2 . The gear ratio is 5 to 1 and the axis of rotation of the engine crank shaft is parallel to that of rear wheel, which has a diameter of 65 cm. Determine the magnitude and direction of the gyroscopic couple when motorcycle rounds a curve of 25 m radius at a speed of 50 km/hr. Total mass of the system is 180 kg and the distance of the combined centre of mass above the ground level when the motor cycle is upright 'h' = 0.6 m. (10)

- VI. (a) What is meant by static balancing and dynamic balancing? (5)
 (b) A, B, C and D are four masses carried by a rotating shaft at a radii 100, 125, 200, and 150 mm respectively. The planes in which the masses revolve are spaced 600 mm apart and the masses of B, C and D are 10 kg, 5 kg and 4 kg respectively. Find the required masses A and the relative angular settings of the four masses so that the shaft shall be in complete balance. (10)

OR

- VII. (a) Explain hammer blow and swaying couple. (5)
 (b) A four cylinder vertical engine has cranks 150 mm long. The planes of rotation of the first, second and fourth cranks are 400 mm, 200 mm and 200 mm respectively from the third crank and their reciprocating masses are 50 kg, 60 kg, and 50 kg respectively. Find the masses of the reciprocating part for the third cylinder and the relative angular positions of the cranks in order that the engine may be in complete primary balance. (10)

- VIII. (a) What is the condition of isochronisms of governors? (5)
 (b) In an engine governor of the porter type, the upper and lower arms are 200 and 250 mm respectively and pivoted on the axis of rotation. The mass of the central load is 15 kg, the mass of each ball is 2 kg and friction of the sleeve together with the resistance of operating gear is equal to a load of 24 N at the sleeve. If the limiting inclinations in the upper arms to the vertical are 30° and 40° , find, taking friction into account, range of speed of the governor. (10)

OR

- IX. (a) Derive an expression for the ratio of belt tensions in a V belt drive. (5)
 (b) A leather belt is required to transmit 7.5 kW from a pulley 1.2 m in diameter running at 250 rpm. The angle embraced is 165° and the coefficient of friction between the belt and the pulley is 0.3. If the safe working stress for the leather belt is 1.5 MPa, density of leather 1 Mg/m^3 and thickness of belt 10 mm, determine the width of the belt taking centrifugal tension into account. (10)
